

Customer Churn Classification

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Machine Learning

Universe

- Best Model: $\{\text{name, accuracy, roc_auc}\}$
- Confusion Matrix: $\{\text{TP, FP, FN, TN}\}$
- Feature Importances: $\{\text{tenure, total_charges, monthly_charges, num_support_tickets, contract_type}\}$
- Cross Validation: $\{\text{n_folds, scores, mean, std}\}$

Variables

Symbol	Meaning	Type / Range
x	decision variable	$\mathbb{R}$

Structure

- **Data**: ~1,000 synthetic customers with 6 features (seed=42)
- **Features**: tenure (months), monthly charges, total charges, support tickets, contract type (month-to-month vs annual), internet service (yes/no)
- **Target**: churned (1) vs retained (0), ~30% churn rate
- **Churn drivers**: shorter tenure, more support tickets, and month-to-month contracts increase churn probability
- **Question**: Which model best predicts churn, and which features matter most?

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

- (1) **Accuracy**: fraction of correct predictions $(\text{TP} + \text{TN}) / \text{total}$
- (2) **Precision**: of those predicted as churned, how many actually churned $(\text{TP} / (\text{TP} + \text{FP}))$
- (3) **Recall**: of those who actually churned, how many were caught $(\text{TP} / (\text{TP} + \text{FN}))$
- (4) **F1 score**: harmonic mean of precision and recall
- (5) **ROC-AUC**: area under the receiver operating characteristic curve; measures discriminative power

- (6) Confusion matrix**: 2x2 table of TP, FP, FN, TN
- (7) Feature importance**: mean decrease in impurity (Gini) across all trees in the Random Forest
- (8) Cross-validation**: 5-fold stratified CV estimates out-of-sample performance and its variability

Approach

Suggested approach Logistic Regression + Random Forest (100) + Gradient Boosting (100)

Complexity class:

Available tools: Logistic Regression + Random Forest

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	No
Algorithm:	Logistic Regression + Random Forest (100) + Gradient Boosting (100)
Time:	0.3412s

Solution Details

Method: 1. Logistic Regression: Linear model with L2 regularization. Fast baseline. Provides interpretable coefficients for each feature.

2. Random Forest (100 trees): Bagged ensemble of decision trees. Handles non-linear relationships and feature interactions. Produces feature importance rankings.

3. Gradient Boosting (100 trees): Sequential boosting of shallow trees. Often achieves the best accuracy on tabular data. Minimizes log-loss via gradient descent in function space.

Verification

✓	Best Accuracy Gt 75Pct	Passed
✓	Best Roc Auc Gt 70Pct	Passed
✓	Rf Outperforms Baseline	Passed
✓	Feature Importances Sum To 1	Passed
✓	Cv Accuracy Stable	Passed
✓	Confusion Matrix Sums Correct	Passed
✓	All Passed	Passed

3. Interpretation

The Question

- **Data**: ~1,000 synthetic customers with 6 features (seed=42)
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- **Target**: churned (1) vs retained (0), ~30% churn rate
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- **Question**: Which model best predicts churn, and which features matter most?

The Answer

A feasible solution was found.

What This Means

- Best model: Logistic Regression (accuracy ~80%, ROC-AUC ~0.85)
- All models beat majority-class baseline (~70% accuracy by always predicting "not churned")
- Top features (Random Forest importance): tenure, total charges, monthly charges, support tickets
- Cross-validation: stable accuracy (std < 0.05 across 5 folds)
- All 7 verification checks pass

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Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

- Solution